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B. E. (Eighth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(ET & T Engg. Branch)

RADAR ENGINEERING and NAVGATIONAL AIDS

(Elective-III)

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory and carries 2 marks and attempt two parts from (b), (c) and (d) and carry 7 marks each.*

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Unit-I

1. (a) What do you mean by a Radar Pulse?

(b) Define Radar Target. Explain Radar cross section of the target.

(c) Explain the integration of Radar Pulse.

(d) Calculate the minimum receivable signal in a radar receiver which has a BW of 1 MHz and noise figure of 6 db.

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Unit-II

2. (a) What do you mean by clutter?

(b) Draw and explain simple block diagram of MTI Radar.

(c) Explain double line canceller with block basic diagram.

(d) A MTI radar is operating at 10 GHz with PRF of 1000 Hz, calculate the lowest three blind speeds.

Unit-III

3. (a) What are basic effect on propagation of Radar waves.

(b) Explain 'Lobbing' effect for RADAR energy scattering from flat earth.

(c) Define duct. Explain the elevated duct and evaporation duct.

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(d) A lower power short range radar with a low noise RF amplifier. Which give an overall noise figure of 4·77 dB. If an antenna diameter is 1 meter, if bandwidth is 500 kHz, the operating frequency is 8

GHz with 5 m^2 cross section are at a maximum distance of 12 km. What must be peak power transmitted.

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Unit-IV

4. (a) What are the different types of Radar Antenna?

(b) Explain the working of parabolic antenna.

(c) Why is a phased array antenna more suitable than a reflector antenna as an ultra low side lobe antenna.

(d) Explain the super heterodyne receiver with the proper block diagram.

Unit-V

5. (a) Define noise figure.

(b) Explain noise figure of radar receiver system.

(c) Explain the duplexes. What is the function of duplexes?

(d) What do you mean by electronics attack and electronic protection.

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